digital pins 0 and 1. This allows us to either communicate from our computer's USB port to the Arduino board or for the Arduino board to communicate with other devices.

Serial communication is a process of sending and receiving bytes of data in a sequential manner. Two of the microcontroller's pins are needed for this communication: pin 0 is the receive pin marked RX and pin 1 is the transmit pin marked TX. These pins are also connected to two of those blinking lights that go mad each time we upload a sketch. When connecting two serial devices to each other, it is necessary to cross-connect the TX pin from Device 1 to the RX pin of Device 2 and vice versa. This also means that anytime we are using the hardware serial communications to talk to things, we cannot have anything else connected to pins 0 and 1. Likewise, if an external device is connected to the Arduino RX or TX pins, then we will not be able to upload a new sketch until those pins have been disconnected.

The serial communications protocol sends data in 8-bit chunks, or 1-byte at a time. In the hardware implementation, the serial bus has an available buffer of 128 bytes, meaning that data up to this size will not be lost if the Arduino is busy doing something else and will be available the next time the data is read. Owing to the usefulness of serial for displaying information in terminal windows and other display devices, there are two ways to send this data: as a numerical value and as a character value. As a byte data type, we can send numerical values between 0 and 255 in each transmitted packet.

Instead of using simple numerical values, we can use the `char` data type to send a character value that corresponds to a standard protocol called **ASCII**, short for American Standard Code for Information and Interchange. ASCII is a method of coding characters with a value from 0 to 255. For example, the character `'a'` translates to the value `97`. We will revisit this shortly but for now, Table 10-1 provides a chart of the ASCII character codes and the corresponding characters.

Table 10-1. *ASCII Character Codes*

Code	Character	Code	Character	Code	Character	Code	Character	Code	Character
32	(sp)	51	3	70	F	89	Y	108	l
33	!	52	4	71	G	90	Z	109	m
34	“	53	5	72	H	91	[	110	n
35	#	54	6	73	I	92	\	111	o
36	$	55	7	74	J	93	]	112	p
37	%	56	8	75	K	94	^	113	q
38	&	57	9	76	L	95	_	114	r
39	‘	58	:	77	M	96	`	115	s
40	(	59	;	78	N	97	a	116	t
41	)	60	<	79	O	98	b	117	u
42	*	61	=	80	P	99	c	118	v